

Others: (6 Marks) Section 1: OS Fundamentals and Concurrent Execution.....Each Question 1/2 Mark

1) The process short-term Scheduler is responsible for managing _____ effectively?	a) <u>CPU</u> b) I/O devices c) Mother Board d) Non-of-the above
2) A Subroutine in the operating system to process a particular type of interrupt is called _____	a) Semaphore b) Dispatcher c) Bootstrap d) <u>Non-of-the-above</u>
3) A _____ move from one state to another state during its life time	a) Swapper b) Pager c) <u>Process</u> d) I-node
4) The one program that is running at all times on the computer is _____	a) <u>Kernel</u> b) Bootstraper c) Semaphore d) Non-of_the_above
5) Internal memory fragmentation	a) occurs in Demand paging technique b) occurs in contiguous memory technique c) decreases memory utilization. d) <u>All-Of-The-Above</u>
6) occurs if a low priority job, though ready, is never scheduled to run because there is always a higher priority job ready.	a) Priority Inversion Problem. b) Multi-levels feedback queues. c) deadlock d) <u>Starvation</u>
7) The busy waiting means:	a) Process blocked waiting for I/O to complete. b) Process spent its quantum for other process. c) Trying to connect with device which doesn't respond. d) <u>None of the above.</u>
8) A semaphore ____.	a) <u>is essentially an integer variable</u> b) is accessed through only one standard operation c) can be modified simultaneously by multiple threads d) cannot be used to control access to a thread's critical sections
9) Blocking implementation of the semaphore mechanism needs_____	a) Swapping b) Paging c) SMP d) <u>waiting queue</u>
10) In Peterson's solution, _____ condition(s) to avoid race conditions are satisfied.	a) one b) tw0 c) <u>three</u> d) four
11) Counting semaphores are used for solving _____.	a) mutual exclusion b) Priority inversion problem c) Starvation d) <u>access control problem to a resource having finite number of instance</u>

12) A deadlocked state occurs whenever ____.	a) a process is waiting for I/O to a device that does not exist b) the system has no available free resources c) <u>every process in a set is waiting for an event that can only be caused by another process in the set</u> d) a process is unable to release its request for a resource after use
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CLO 6/SLO I: (4 Marks) Section 2: Memory Management

Each Question 1/2 Mark

13) The address of a segmentation table in memory is pointed by _____	a) stack pointer b) page table base register c) page register d) <u>None of the above</u>
14) In contiguous memory allocation with fixed sized partition, the degree of multi programming is bounded by _____	a) <u>the number of partitions</u> b) the CPU utilization c) the memory size d) all of the above
15) Given the reference string of page accesses: 1 2 3 4 2 3 4 1 2 1 1 and a system with three page frames, what is the final configuration of the three frames after the true LRU algorithm is applied?	a) 2, 1, 3 b) <u>4, 1, 2</u> c) 3, 4, 1 d) 1, 2, 3
16) In a 16-bit machine with page size 1K byte and configured with maximum adequate frames. Which program size is impossible to load into memory managed by paging technique?	a) 16 KB b) 5.5 KB c) 32000 Bytes d) <u>68 KB</u>
17) What is compaction?	a) a technique for overcoming internal fragmentation b) a paging technique c) <u>a technique for overcoming external fragmentation</u> d) a technique for overcoming fatal error
18) In a 16-bit machine with 4k page size. What is the maximum number of pages that can be addressed in this system?	a) 16 b) 4 c) <u>8</u> d) 32
19) If a page number is not found in the TLB, then it is known as a :	a) <u>TLB miss</u> b) Buffer miss c) TLB hit d) All of the mentioned
20) In Paging Technique every address generated by the CPU is divided into two parts :	a) frame bit & page number b) <u>page number & page offset</u>

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| | c) page offset & frame bit
d) frame offset & page offset |
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CLO 8/SLO I: (Marks 3) Section 3: Disk Fundamentals

Each Question has 1/2 Mark

21) File Control Block contains the file size beside other file attributes.	T
22) Disk Linked allocation method ensures that only one access is needed to get a disk block using direct access.	F
23) File whose blocks were allocated using Indexed allocation method needs 2 disk block access in order to read or write any byte of the file.	T
24) On 32-bit machine a FAT file system can't be used to format a terabyte hard disk as one partition.	T
25) Which of the following cannot be considered as disk access delay?	a) transfer delay b) rotational delay c) seek delay d) <u>cache memory access delay.</u>
26) Disk file system with contiguous allocation is suitable for formatting?	a) <u>CDROM</u> b) Video tape c) Hard Disk d) Flash memory sticks

Others: (6 Marks) Section 4: Process Scheduling

(Each question 1.5 Mark)

27) What is an Address binding means?



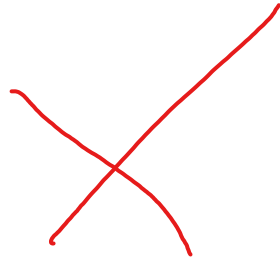
28) Mention 2 process states and explain how to enter and leave such 2 states.



29) What would happen if round robin quantum was chosen too long, and what would happen if it was too short?

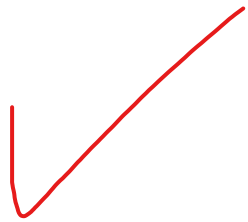


30) Mention four CPU scheduling events when scheduling can occur during the life cycle of a process?



CLO 6/ SLO I: (Marks 5) Section 5: Memory Management
Each has 1 Marks

31) Virtual memory allows a process to execute partially in memory when?



32) Explain the phenomena of process trashing?



33) State the advantages and disadvantages of pre-paging and demand paging memory management techniques?



34) In a paging system assume it takes 20 nsec to search TLB and 100 nsec to access memory. 98% of the time we are lucky and get the pages numbers in the TLB. Calculate the effective memory access time.

$$.98 \times 20 + 100 + 0.02 \times 20 + 100 + 100$$

$$P(\text{hit}) \times T(\text{hit}) + P(\text{miss}) \times T(\text{miss})$$

35) A pure paging system uses 16-bit address and 1K pages. The following shows the page tables of Process 1. Translate the logical address 2048 of Process Fill your answers into the table below.

[1 Marks]

Process 1

0	2
4	3
5	8
2	0

$$2048 / 1024 = 2.0$$

Process	Address	Page number	offset	Physical Address
Process 1	2048	5	0	$8 * 1024$

CLO 8/ SLO I: (Marks 6) Section 6: Disk Management

36) Mention 2 Components of the File System?

[1Marks]



- 37) Hard disk block size is an important issue in file system design. What are the advantages and disadvantages of choosing large or small block size [1 Marks]

- 38) Assume that the block size is 1024 bytes, fill in the following spaces [2 Marks]

1. The physical Address of byte number 1045 in a file stored using contiguous allocation is _____. Keep in mind that this file is stored in several block starting from block number 19 and end at block 24.

2. The physical Address of byte number 1045 in a file stored using the following FAT is _____. Keep in mind that the starting entry for this file is block 4.

		8		7			2	-1	
0	1	2	3	4	5	6	7	8	9

- 39) For A Linux file system with i-node of 13 pointers and uses a block of size 1 Kbytes, consider a file of size 11.5 Kbytes. For such file, fill in the space in the following:

[2 Marks]

- i. How many sectors are allocated for such file if sector size is 512 bytes?
ii. How many disk access do you need to read byte number 2010? 1

$$(11.5 / 1) = 12 \quad \text{first 11 direct access is 1 Kbytes.}$$
$$12 \times \frac{1024 \text{ bytes}}{512 \text{ bytes}} = 24 \text{ Sectors}$$